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Mental design and (second) language epistemology: adjectival restrictions of *wh*-quantifiers and tense in English–French interlanguage

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This study addresses the issue of second language (L2) epistemology assuming Chomsky's (1995) discussion of the place of Universal Grammar (UG) in mental design: i.e., the optimal solution to the mental design problem for language in the sense of Minimalist theory. Aspects of interpretation of continuous and discontinuous interrogatives of the form *qui de AP* ('who (of) AP') in first language and L2 acquisition appear to follow from principles of economy in mental design and language-dependent hypotheses. We argue that such knowledge is guaranteed to arise in the absence of relevant input only if a grammar is a realization of language-dependent hypotheses and basic principles of grammar, but crucially not if it consists of a set of (derivative) grammatical theorems not constrained by principles of optimal design.

I Mental design and second language epistemology

Within generative approaches to second language (L2) acquisition, a number of researchers have offered versions of the Strong UG Hypothesis, the claim that the full set of Universal Grammar (UG) principles constraining native language (L1) acquisition also constrains (adult) L2 acquisition (White, 1985a; 1985b; 1989; duPlessis *et al.*, 1987; Schwartz and Tomaselli, 1990; Martohardjono, 1993; Thomas, 1993; Schwartz and Sprouse, 1994; Vainikka and Young-Scholten, 1994; Epstein *et al.*, 1996; Dekydtspotter *et al.*, 1997; Kanno, 1998 and many others). Nevertheless, many linguists would be hesitant to accept the Strong UG Hypothesis, largely on the basis of the non-target performance of most adult L2 acquirers (e.g., Newmeyer, 1998). A body of research indeed concludes that UG (in any shape or guise) plays no role whatsoever in (adult) L2 development (Felix, 1985; Clahsen and Muysken, 1986; Meisel,

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1997), while others contend that UG effects on interlanguage (IL) knowledge are at best partial (Zobl, 1986; Clahsen and Muysken, 1989; Bley-Vroman, 1990; Tsimpli and Roussou, 1991; Strozer, 1994). However, as Schwartz (1987) argues, if the treatment of linguistic input necessitates a domain-specific language processing module, then adult L2 acquisition should be domain-specific in character.

Recent work in Chomsky's (1993; 1995; 2000) Minimalist Program calls for a very particular characterization of mental design associated with human (L1) linguistic knowledge. On this view, grammatical architecture constitutes an optimal design solution to the problem that a language system poses for mental architecture. As Chomsky (1995: 1) notes:

This work [The Minimalist Program] is motivated by two related questions: (1) what are the general conditions that the human language faculty should be expected to satisfy? and (2) to what extent is the language faculty determined by these conditions, without special structure that lies beyond them?

Thus, in addition to the lexicon, phonological (PF) and conceptual (LF) representations and a derivational algorithm (C_{HL}) linking PF and LF are necessary and sufficient components of language. They are the only components of grammar. C_{HL} is subject to principles of economy: binary Merge is the minimal operation that will allow concatenation (for discussion, see Collins, 1997; Epstein *et al.*, 1998). Derivations must admit of just the necessary operations and derived objects: no operation exists unless it is required (Last Resort, Greed, Procrastinate) and unless it is the smallest such operation possible (Minimal Link, Shortest Move, Fewest Operations). No C_{HL} object can exist unless it is legible at either interface (Full Interpretation). On this view, where principles of grammatical architecture have this level of generality, IL knowledge should bear precisely these reflexes.

We show that, on this view of grammar, the interpretation of adjectival restrictions in interrogative expressions of the form *qui* (...) *de AP* ('who (of) AP') follows elegantly from the nature of the human language computational system. We also show that English–French IL knowledge of the interpretation of such adjectival restrictions of *wh*-quantifiers seems to require such an organizational structure where a language-particular lexicon interacts with a universal computational system (C_{HL}), obeying general principles of mental design of the type investigated by Minimalism.

The paper is organized as follows: Section II discusses the interaction of the adjectival restriction to interrogative expressions

of the form *qui* (...) *de AP* ('who (of) AP') with tense in Minimalist design. Section III contrasts the plausibility of nativist and non-nativist approaches to the etiology of such grammatical knowledge. Section IV reports the results of an empirical study of sensitivity to these interactions on the part of adult English–French acquirers. Section V argues that the empirical evidence is most compatible with the Minimalist design assumptions considered above. We conclude that this strengthens an emergent body of research documenting reflexes of mental design in L2 acquisition.

II Principles of economy, adjectival restrictions of *wh*-quantifiers, and tense

In French the interrogative pronoun *qui* ('who') is found either together with its adjectival restriction *de célèbre* ('of famous'), as in (1), or separated from its adjectival restriction, as in (2). We refer to (1) and (2) as continuous and discontinuous interrogatives, respectively (for detailed discussion, see Huot, 1981; Azoulay-Vicente, 1985; Birdsong, 1985; Hulk and Verheugd, 1994).

- 1) Qui de célèbre fumait au bistro dans les années 60?
 who of famous smoked in-the bar in the '60s
- 2) Qui fumait de célèbre au bistro dans les années 60?
 who smoked of famous in-the bar in the '60s
 'Which famous person smoked in bars in the '60s?'

The structures in (1) and (2) are not semantically equivalent. The interrogative in (1) allows an interpretation of *célèbre* as either the (set of) individuals famous at the speech time or the (set of) individuals famous at the past interval of the 1960s, whereas the interrogative in (2) allows a interpretation of *célèbre* only as the (set of) individuals famous at the past interval of the 1960s, but not as the (set of) individuals famous at the speech time.

Thus, suppose that in the 1960s Herman the Hermit was a famous rock star and was often seen at bars smoking with Linda Tripp, who was then totally unknown; whereas now it is Linda Tripp who is famous (at least, she was when we conducted the experiment reported here), and neither of them smokes any more. In this situation, the question in (1) can be properly answered with Herman the Hermit or with Linda Tripp. However, the question in (2) can be answered only with Herman the Hermit, and crucially not with Linda Tripp. Thus, we are concerned with the fact that in (1) *célèbre* allows both a past and a speech time answer, whereas in (2) *célèbre* must receive a past time interpretation, but

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cannot be construed at the speech time.

The syntax and interpretation of French continuous vs. discontinuous interrogatives can be understood in terms of the theory of movements and principles of economy. Following Chomsky (1995), movement is triggered by the need to satisfy the ‘attraction’ requirements of certain functional heads. We assume with Hulk and Verheugd (1994) that *qui* in expressions such as *qui de célèbre* is the specifier of a predication functional projection with *célèbre* as complement. Assuming the VP-internal subject hypothesis, the interrogative in (1) has the syntactic structure in (3), where in the derivation the trace $t_{\text{qui de célèbre}}$ is simply a copy of the moved item. The movement of *qui de célèbre* is driven by the need to satisfy a strong *wh*-feature of the interrogative C.

- 3) [_{CP} Qui de célèbre [_C [_{TP} $t_{\text{qui de célèbre}}$ fumait [_{VP} $t_{\text{qui de célèbre}}$ [_{V'} t_{fumait}]
 who of famous smoked
 au bistro]]]]?
 at-the bar
 ‘Which famous person smoked in bars?’

We assume with Chomsky (1995: 265) that, when required by the sensorimotor system, a categorial chain CH_{CAT} is constructed alongside the formal feature chain CH_{FF} consisting of the set of formal features and its trace. A strong feature requires the construction of CH_{FF} . If $\text{FF}(\text{wh})$, the set of formal features including the [*wh*] feature, corresponds to the whole expression *qui de célèbre*, then Full Interpretation requires that a categorial chain CH_{CAT} be constructed such that the phonological features of the expression *qui de célèbre* be delivered to the phonology (i.e., all features relevant to the phonological level must be interpreted). Thus, (3) results in the continuous utterance.

Given the possibility of left-branch extraction in French, the very existence of *qui de célèbre* suggests that the derivation in (4) and its interpretation must be taken into consideration. The feature-checking requirements of I and C are potentially satisfied by the successive movement of *qui* (4). Thus (4) constitutes a left-branch extraction along the lines of *combien* extractions (Hulk and Verheugd, 1994).¹

¹ Hulk and Verheugd (1994) assume that (2) is structurally ambiguous between (i) and (ii), where *de célèbre* is generated directly. What is crucial for our argument is that the introduction of the modifier on a cycle determines the interpretation on that cycle.

- i) [_{CP} Qui [_C [_{IP} t_{qui} [_{I'} fumait [_{VP} [t_{qui} de célèbre] [_{V'} t_{fumait}] au bistro]]]]?
 who smoked of famous at-the bar
 ii) [_{CP} Qui [_C [_{IP} t_{qui} [_{I'} fumait [_{VP} [_{VP} [t_{qui} [_{V'} t_{fumait}] de célèbre] au bistro]]]]?
 who smoked of famous at-the bar

- 4) [_{CP} Qui [_C [_{IP} *t*_{qui} [_{I'} fumait [_{VP} [*t*_{qui} de célèbre] [_{V'} *t*_{fumait}]
 who smoked of famous
 au bistro]]]]?
 at-the bar

Once *qui* has raised to SpecI and SpecC as in (4), the features of I and C triggering movement are satisfied (and deleted under the virus theory). Neither I nor C is an attractor for $[t_{\text{qui}} \text{ de célèbre}]$. The resulting putative movement would violate Last Resort, since the operation is not motivated by a (live) checking relation. Thus, (4) results in a word order as in (2).

Following Epstein *et al.* (1998), we assume that the relevant aspects of the expression *qui de célèbre* are interpretable at various stages of the derivation. LF-‘legibility’ can be construed as the derivability of a logical/conceptual representation based on the logical representations of lexical items and LF syntactic representations, while PF-‘legibility’ can be construed as the derivability of a morphophonological representation satisfying the relationships of the (functional) lexicon.

The interrogative value of a sentence is provided by a question operator **Q** introduced at the CP level in the LF representation of the interrogative sentence. Following Groenendijk and Stokhof (1982), we assume that $[_{CP} \textbf{Q} [_{IP} \textbf{John left}]]$ denotes a function that maps any world w to true if the state of affairs described by **John left** holds both of the real world and of w and to false otherwise. This function partitions the worlds into those where John left and those where he did not. Answering John left (or yes) characterizes the real world as being in the first set. Constituent questions involve the question operator (**Q**) and *wh*-features (**wh**). The semantic contribution of **wh** is that of binder. **Wh** binds a *wh*-variable x introduced by a *wh*-expression (**wh** x). Thus, the binding configuration $[_{CP} \textbf{wh } x [_{IP} \textbf{Pred}(x)]]$ derives in the semantics a function that, at some world, w carves out the set of individuals satisfying the predicate description at w . The LF structure $[_{CP} \textbf{wh } x \textbf{Q} [_{IP} \textbf{Pred}(x)]]$ derives in the semantics a function that for any world w maps w to true, if the individuals satisfying the predicate description at w are the individuals that satisfy the predicate description at the real world w_o , and to false otherwise. Thus, answers to *wh*-questions provide the individuals satisfying the IP description at w_o .

The *wh*-feature of *qui de célèbre* requires a CP-related interpretation with the *wh*-feature of C. The adjectival description can enter the logical representation at the CP level (5), where it restricts the binder, or the variable at the IP level (6) or at the VP

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level (7), where it restricts the variable. The LF expression in (5) determines a restricted binder $[_{CP} \textbf{wh } x \textbf{ célèbre}(x) [_{IP} \textbf{Pred}(x)]]$ deriving a partial function that, at some world w , carves out from the famous individuals at w those that satisfy the predicate description at w . In (6) and (7), the binder is not restricted (a set of famous individuals is not pre-established); *célèbre* is part of the descriptive contents of the function. The descriptive contents of the function (and thus the answerability conditions) differ with respect to whether *célèbre* is within the scope of tense represented by the past operator notated **P** in the logical representation.

5) $[_{CP} \textbf{wh } x \textbf{ célèbre}(x) \textbf{Q} [_{IP} \textbf{P} [_{VP} \textbf{fumer-au-bistro}(x)]]]]$

6) $[_{CP} \textbf{wh } x \textbf{Q} [_{IP} \textbf{célèbre}(x) \& [_{I'} \textbf{P} [_{VP'} \textbf{fumer-au-bistro}(x)]]]]]]$

7) $[_{CP} \textbf{wh } x \textbf{Q} [_{IP} \textbf{P} [_{VP} \textbf{célèbre}(x) \& \textbf{fumer-au-bistro}(x)]]]]$

In (5), the question can be paraphrased as *Gi en famous people, who smoked in bars?* and is relevant only of the famous people in the world. In (6), the question is *Who smoked in bars and is now famous?* This is because in (6) *célèbre*(x) is not within the scope of **P** determined by the VP expression. In (7), the individuals were famous and smoked in bars at some past interval. This is because in (7) *célèbre*(x) is part of the VP expression and thus within the scope of **P**.

Once *qui* has raised to SpecI and SpecC as in (4), I or C are not attractors for $[t_{\text{qui}} \text{ de célèbre}]$. Interpretations associated with raising $[t_{\text{qui}} \text{ de célèbre}]$ would violate Last Resort. The derivation in (4) requires the interpretation in (7).

To recapitulate, Last Resort (requiring that operations be externally motivated) and Full Interpretation (requiring that representations be externally motivated) leave (7) as the only possible logical form for the string in (2). Once the movement of *qui* satisfies the *wh*- and EPP-features of C and I as in (4), C and I are no longer available targets for further operations. Hence, the substitution of the *wh*-expression $[t_{\text{qui}} \text{ de célèbre}]$ for *qui* violates Greed. If *qui de célèbre* enters in a checking configuration with C at the PF-interface, then Full Interpretation at PF requires that *qui de célèbre* be uttered. This has the effect of ensuring that if *qui* is raised, the restriction is interpreted under tense; whereas if *qui de célèbre* is raised, the interpretation of the restriction is interpreted under or over tense. The derivation of *qui de AP* interrogatives requires only a morphological trigger and is non-defeasible by potential noise. Knowledge of (the interpretation of) French

sentences can be profitably viewed as the result of a French-dependent functional lexicon (which can be thought of as a set of hypotheses about the input) and a universal deductive procedure (C_{HL}), obeying general principles of mental design of the type investigated by Minimalism.

III Learnability

While English-speaking classroom learners of French may be exposed to discontinuous and continuous interrogatives in naturalistic conversations, learners receive no instruction on the existence of discontinuous interrogatives, and the interpretive characteristics of discontinuous and continuous interrogatives are discussed neither in pedagogical grammars of French nor in classroom presentations. The vast majority of French language instructors are not consciously aware of the properties under investigation. Below we present experimental evidence suggesting that French native speakers typically find speech time interpretations even of continuous interrogatives difficult. It is thus very unlikely that learners will receive anything like robust exposure to <continuous interrogative, speech time answer> pairs.

In this section, we cast severe doubts on the plausibility of accounting for such knowledge on the basis of induction. If L2 learners rely on reorganization of surface strings, as Meisel (1997) argues, the task of the learner is to determine from exposure alone that continuous and discontinuous interrogatives are not simple permutation variants. The learner presumably assumes that a sentence is uttered only in a context that supports it (at least as the speaker sees it, because the speaker is assumed to act in good faith following Gricean maxims). On such an assumption, determining that discontinuous interrogatives are not permutations variants of continuous interrogatives would require a situation in which the discontinuous interrogative is felicitous and the continuous interrogative infelicitous. If the learner assumes that the continuous word order is basic, he or she faces an insuperable problem. This is because the discontinuous interrogative in (2) cannot express every question that is expressible by the continuous interrogative in (1), but the continuous interrogative in (1) can express every question expressible by the discontinuous interrogative in (2). Thus, the set of possible interpretations of the interrogative in (2) is a proper subset of the set of possible interpretations of the interrogative in (1). Whenever the interrogative in (2) can be asked and answered in a particular way, the interrogative in (1) can be asked and answered in the same way as well. Hence, no situation of use can

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disabuse the learner of the belief that (2) is a synonymous rewrite variant of (1).

Even if the learner does not assume that (2) is a rewrite variant of (1), he or she still faces enormous difficulties. The learner must come to the learning task expecting both a class of readings with scopal dependence of the adjectival restriction vis-à-vis the tense operator and a class of readings with scopal independence of the adjectival restriction vis-à-vis the tense operator. Learning French involves determining which readings are expressible by a given syntactic form and which are not: i.e., matching sentence structures with expected interpretations. There must be (idiosyncratic) constraints on the hypotheses that are generated, since over-generation could soon make the learning task computationally unrealistic. Idealized input would contain evidence that continuous interrogatives allow the full range of expected interpretations, but no evidence could be found to support the proposition that discontinuous interrogatives allow scope of the adjectival restriction both over and under the tense operator. Given the pattern in the input, the learner would (putatively) infer that there is no interpretation of the adjectival restrictions at the speech time in discontinuous interrogatives.

It is not clear how the process of normal communication could lead to acquisition of the relevant property. A scenario of the type described in the previous paragraph presupposes that the learner is an observer of a situation where a question is asked and an answer provided, and this answer is subject to independent confirmation. However, the presence of confirming evidence is in most cases lacking. In the usual case, humans utter interrogatives in order to gain information they do not already have. Furthermore, in the usual case, when a conversation partner offers a response, both the questioner and the answerer assume that the latter has interpreted the interrogative as the former intended it and has responded to it accurately and honestly (Grice, 1975).

Suppose that the learner were to mistakenly use a discontinuous interrogative expecting a (positive) identification of an individual presently famous as an answer but the answer was given ostensibly on the basis of past famous individuals. Even then, the learner might infer that the answerer's knowledge of the world is flawed or that the answerer intends to deceive. Thus, even such an opportunity would not reliably lead to corrective feedback. Furthermore, the inference that discontinuous interrogatives may not permit the interpretation of the adjectival restriction outside the scope of tense is potentially refutable: it can be defeated by a counterexample, even if merely 'noise'. Such a counterexample would be provided

by the *erroneous* belief that an answer given to a discontinuous interrogative refers to an individual who is currently famous (hence, is associated with a construal of the adjectival restriction over the scope of tense), contrary to what the answerer intends. This is in sharp contrast to the hypothesis that knowledge of (the interpretation of) French interrogatives is deductively determined by a French-dependent function lexicon and a universal deductive procedure (C_{HL}).

Turning to knowledge of English, the English glosses of the French sentences in (8) and (9) are ungrammatical, as are the 'adjusted' glosses in (10) and (11), where the word *of* is omitted. Despite their ungrammaticality, the phrase (*of*) *famous* seems plausibly understood as restricting the *wh*-quantifier *who*.

- 8) * Who of famous smoked in bars in the '60s?
(Qui de célèbre fumait au bistro dans les années 60?)
- 9) * Who smoked of famous in bars in the '60s?
(Qui fumait de célèbre au bistro dans les années 60?)
- 10) ?* Who famous smoked in bars in the '60s?
(Qui de célèbre fumait au bistro dans les années 60?)
- 11) * Who smoked famous in bars in the '60s?
(Qui fumait de célèbre au bistro dans les années 60?)

The ungrammaticality in English resides in the fact that the English functional lexicon (lacking a correspondent to French *de*, 'of', in this case) fails to license the parse of the input in (8)–(11). At this point, it is very important to note that the failure of the English lexicon to license the parse in this modular system does not entail the failure of the other systems of knowledge, such as C_{HL} . Sentences with parses not licensed by (i.e., violating) the parameter values of the English functional lexicon are nevertheless C_{HL} -derivable and LF-interpretable (on some other parameter value). Given the modular organization of linguistic knowledge, it is proper to ask what denotations such ungrammatical sentences may and may not receive. We must also ask whether speakers can reliably *intuit* what a given ungrammatical sentence may and may not mean. Intuitive interpretability as a performance notion seems in part a function of the strength of the syntactic violation. Evidence of relevant asymmetries on a task would confirm the hypothesis that these patterns arise in intuitions solely by virtue of 'general principles rooted in the specific character of the human mind' (Chomsky, 1972: 102), since ungrammatical sentences are

presumably absent in the input. We address the issue of intuitive interpretability experimentally.

To recapitulate, the (un)grammaticality of discontinuous interrogatives in English vs. French is due to variation in the (language-dependent) functional lexicon; whereas interpretability, even if not manifested at the level of performance, is ensured for analyses satisfying the requirements of the computational system of natural language grammars (C_{HL}). If asymmetries arise in IL, beyond the intuitive interpretability levels that knowledge of English is able to provide, then such IL development must arise as deductive consequences of a more French-like IL functional lexicon in the human-language computational system. This is because only domain-specific deduction of the type that we have considered seems to *guarantee* acquisition across levels of exposure.

IV The study

1 *Experimental procedures*

Our experiment investigated two syntactic structures, continuous and discontinuous interrogatives, and two kinds of interpretations, an interpretation involving the evaluation of the restriction at a past time reference and an interpretation involving the evaluation of the restriction at the speech time. In the Target Language (TL), continuous interrogatives are potentially associated with past time and speech time evaluations of the restriction, but discontinuous interrogatives can be paired only with the past time evaluations of the restriction. Crucially, discontinuous interrogatives cannot be paired with a speech time evaluation of the restriction (where a (non-present) tense operator is present).

Each item in the experimental instrument consisted of (1) a prose narrative, (2) a question about the prose narrative and (3) an answer to that question. Respondents were to indicate whether the answer was a correct answer. Experimental items included 8 sets of 4 sentences (quadruples). Within each quadruple, continuous vs. discontinuous interrogatives were crossed with answers corresponding to the speech time vs. the past time evaluation of the restriction. Each narrative provided both potential speech time and past time answers. Hence, the contexts were neutral, so that in addition to probing for asymmetries between discontinuous vs. continuous interrogatives given to speech time and past time evaluation of the restriction, the instrument could also measure the relative accessibility of speech time vs. past time answers to discontinuous and continuous interrogatives. The lexical material in

the <interrogative, answer> pairs within a given quadruple was identical and was limited to vocabulary familiar to respondents even at the most elementary level investigated. For each quadruple, we composed two brief prose narratives in English: one unambiguously involving a scenario matching a speech time answer and another unambiguously involving a scenario matching a past time answer. Each narrative then appeared twice: once paired with a sentence exemplifying a continuous interrogative and once paired with a sentence exemplifying a discontinuous interrogative.

We illustrate the design of a quadruple with an actual example from the task. (Appendix 1 provides a summary of all eight experimental quadruples.) In this experiment, a character – a teacher of English (Mme Goyette) – asks a comprehension question about a story presented in English, and a second character – a pupil (*élève*) – responds to Mme Goyette's question. The role of the respondent is to determine whether the pupil provided a correct answer to this question. Respondents knew that, although several answers may (in principle) be possible, they had to concentrate on the answer given by the pupil (see Appendix 2). The questions in Sample Test Items 1–4 (see Figures 1–4) ask: 'Which famous person smoked in the bar in the 1960s?' Sample Test Item 1 offers a past time answer paired with a continuous interrogative. The answer *Herman the Hermit* corresponds to the past time evaluation of the adjectival restriction. (The speech time answer would have been *Linda Tripp*, since *Linda Tripp* is now famous.) Sample Test Item 2 includes the same past time answer, but this time paired with a discontinuous interrogative. Sample Test Item 3 offers a present time answer and a continuous interrogative. Here the answer

Attitudes toward smoking have changed drastically since the 1960s. In the 1960s many people would go to bars and smoke every night. For example, Herman the Hermit was a famous rock star in those days and was often seen at bars smoking with Linda Tripp, who was then totally unknown. How times have changed! It is Linda Tripp who is famous, and neither of them smokes any more.

Mme Goyette: Qui de célèbre fumait au bistro pendant les années 60?

Élève: Herman the Hermit

IS THIS A CORRECT ANSWER TO THIS QUESTION?

Figure 1 Sample Test Item 1: Past time answer paired with continuous interrogative

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Attitudes toward smoking have changed drastically since the 1960s. In the 1960s many people would go to bars and smoke every night. For example, Herman the Hermit was a famous rock star in those days and was often seen at bars smoking with Linda Tripp, who was then totally unknown. How times have changed! It is Linda Tripp who is famous, and neither of them smokes any more.

Mme Goyette: Qui fumait de célèbre au bistro pendant les années 60?

Élève: Herman the Hermit

IS THIS A CORRECT ANSWER TO THIS QUESTION?

Figure 2 Sample Test Item 2: Past time answer paired with discontinuous interrogative

Attitudes toward smoking have changed drastically since the 1960s. In the 1960s many people would go to bars and smoke every night. For example, Herman the Hermit was a famous rock star in those days and was often seen at bars smoking with Linda Tripp, who was then totally unknown. How times have changed! It is Linda Tripp who is famous, and neither of them smokes any more.

Mme Goyette: Qui de célèbre fumait au bistro pendant les années 60?

Élève: Linda Tripp

IS THIS A CORRECT ANSWER TO THIS QUESTION?

Figure 3 Sample Test Item 3: Speech time answer paired with continuous interrogative

Attitudes toward smoking have changed drastically since the 1960s. In the 1960s many people would go to bars and smoke every night. For example, Herman the Hermit was a famous rock star in those days and was often seen at bars smoking with Linda Tripp, who was then totally unknown. How times have changed! It is Linda Tripp who is famous, and neither of them smokes any more.

Mme Goyette: Qui fumait de célèbre au bistro pendant les années 60?

Élève: Linda Tripp

IS THIS A CORRECT ANSWER TO THIS QUESTION?

Figure 4 Sample Test Item 4: Speech time answer paired with discontinuous interrogative

Linda Tripp corresponds to a past evaluation of the adjectival restriction *de célèbre*. (The past time answer would have been *Herman the Hermit*, since that is who was famous in the 1960s.) Sample Test Item 4 has the same speech time answer, now paired with a discontinuous interrogative.

We also included 18 fillers designed to blend in with the rest of the task, as well as to offset, at least in part, the expected imbalance between the positive and negative responses. Thus, there were 32 experimental items plus 18 fillers, for a total of 50 items. The 50 items were presented in three distinct random orders.

Since the meaning of a question is tied to the (truthful) answers to it, respondents were given question–answer pairs in French and asked whether the answer is a correct answer to the question. We presented the narratives in English to ensure that all of the respondents in the study group clearly and unambiguously understood the context in which the putative sentence is uttered. Restricting the use of the TL to the sentences to be judged also allowed us to study learners at low levels of proficiency, who might not be able to deal with an entire context in the TL. Finally, presenting only the test sentences in the TL eliminated the possibility that the respondents might somehow use the surface grammatical form of the contexts as an aid in judging the sentences. Presenting a context in English followed by a test sentence in the TL might introduce the confounding factor of bilingual processing into the task. However, this would not skew the results in a way that would be relevant to our study, since it would presumably play the same role in judging both the acceptable and unacceptable pairings in each of the distinctions we investigated. A possible effect of test sentences that are unacceptable in English would be a depression of acceptance rates, but this effect would then be evident for both scenarios with which such a sentence is paired.

Our respondents included a group of 47 intermediate classroom learners of French from students enrolled in third-semester and fifth-semester French language-proficiency courses at Indiana University and a group of 11 advanced English–French IL speakers, who had lived for at least one year in a French-speaking country in the course of earning a university degree in French studies. The volunteers provided us with background information on their L1, their exposure to the TL, and any other languages they had studied or learned. We excluded from consideration volunteers whose L1 was not English, and we controlled for the effect of prior exposure to languages other than English and the TL. Our ‘control’ groups included 30 native French students enrolled in English language-proficiency courses at Université de Pau et Pays de L’Adour and

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47 native speakers of English who were enrolled in German language-proficiency courses at Indiana University and who had never received instruction in French.

2 *Results*

The intermediate learners ($n = 47$) accepted speech time answers to discontinuous interrogatives at the rate of 25.00%; whereas they accepted speech time answers to continuous interrogatives at the rate of 41.22%. (See Appendix 3, Tables 1d and 2d for item-by-item tabulations of results on continuous and discontinuous interrogatives, respectively.) This is statistically significant ($t(46) = 3.36, p = .002$). This much lower rate of acceptance of speech time answers to discontinuous interrogatives does not appear to be due to a rejection of an odd surface pattern but due to the interaction of a syntactic pattern with a semantic representation. This is because there was no significant difference in respondents' acceptance of past time: answers both discontinuous and continuous interrogatives were accepted at 90.69% ($t(46) = .00, p = 1.00$); see Table 1.²

The advanced learners ($n = 11$)³ showed similar (but more pronounced) patterns (see Table 2), accepting speech time answers to discontinuous interrogatives at the rate of 15.90%, significantly

Table 1 English–French interlanguage (intermediate learners) ($n = 47$): percentage 'yes' on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech interpretation	Past interpretation
Continuous	41.22 (34.47)	90.69 (21.08)
Discontinuous	25.00 (24.03)	90.69 (15.30)

² A similar response pattern is observable on the basis of 'no' responses which are almost linearly dependent, and therefore virtual repeated measures (see Appendix 4, Tables 1a–5a for 'no' responses corresponding to the 'yes' responses reported in Table 1–5, respectively, in the text). Thus, intermediate learners ($n = 47$) rejected answers corresponding to the speech time evaluation of discontinuous adjectival restrictions at the rate of 73.40%; whereas they rejected answers corresponding to the speech time evaluation of continuous adjectival restrictions at the rate of 56.75%. Again, this difference of means achieves statistical significance ($t(46) = 3.51, p = .001$). This is not what is expected merely as a familiarity effect with the syntactic pattern since there was no statistically significant difference in learners' rejection of answers corresponding to the past time evaluation of discontinuous adjectival restrictions (9.57%) and learners' rejection of answers corresponding to the past time evaluation of continuous adjectival restrictions (7.98%) ($t(46) = .45, p = .658$).

³ An anonymous reviewer notes the large difference in learner group sizes. What is crucial here is that the sample size suffices to capture contrasts in a reliable manner given the means obtained.

Table 2 English–French interlanguage (advanced learners) ($n = 11$): percentage ‘yes’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation	Past time interpretation
Continuous	46.59 (37.95)	79.55 (26.96)
Discontinuous	15.90 (19.43)	90.90 (14.88)

less than speech time answers to continuous interrogatives at the rate of 46.59% ($t(10) = 2.59$, $p = .027$). In contrast there was no statistically significant difference in their acceptance of past time answers to discontinuous interrogatives (90.90%) and to continuous interrogatives (79.55%) ($t(10) = 1.12$, $p = .290$).⁴ A French native speaker comparison group provides experimental indication that the input received by learners provides poor evidence for the restrictions on the interpretations of detached adjectival restrictions (see Table 3).

An asymmetry in the relative acceptance of speech time answers to discontinuous *qui de* interrogatives vs. continuous *qui de* interrogatives also surfaces in the native group ($n = 30$). Thus, considering acceptance of speech-time answers, we find an acceptance rate of 5.00% for discontinuous interrogatives vs. 12.50% for continuous interrogatives. Because of a large standard deviation (SD = 27.46) on speech time answers to continuous interrogatives, the two means do not achieve statistical significance ($t(29) = 1.61$, $p = .119$). Nevertheless, this difference in the means is

Table 3 Native French group ($n = 30$): percentage ‘yes’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation	Past time interpretation
Continuous	12.50 (27.46)	88.75 (20.06)
Discontinuous	5.00 (7.76)	96.25 (9.93)

⁴ A similar response pattern is observable on the basis of ‘no’ responses (see Appendix 4): the Advanced learner group ($n = 11$) rejected answers corresponding to the speech time evaluation of discontinuous adjectival restrictions at the rate of 80.68%; whereas they rejected answers corresponding to the speech time valuation of continuous adjectival restrictions at the rate of 48.45%. This difference of means is significant ($t(19) = 2.44$, $p = .035$). They rejected answers corresponding to the past time evaluation of discontinuous adjectival restrictions at the rate of (5.68%) and answers corresponding to the past time evaluation of continuous adjectival restrictions at the rate of (13.63%) ($t(10) = .92$, $p = .379$).

theoretically significant.⁵ For past time answers, the difference in the acceptance between discontinuous interrogatives (96.25%) vs. continuous interrogatives (88.75%) reaches significance ($t(29) = 2.16, p = 0.037$).

Hence, native speakers have difficulty accessing the speech time answers even where they are syntactically licensed. Therefore, it is unlikely that the input could be rich enough to reliably (i.e., measurably) inform learners that speech time answers to discontinuous interrogatives are impossible, whereas speech time answers to continuous interrogatives are possible. Perhaps the preference for past tense evaluations of the restriction could ultimately be captured in terms of a serial parser and reconstruction (minimization of the restriction) as a reflex of the creation of operator-variable structures (for discussion, see Chomsky, 1995; Epstein *et al.*, 1998).

We now present evidence that the results on the task could not be obtained on the basis of English judgements, via a ‘glossing’ strategy. To replicate what glossing would be like, we transcribed word by word the experimental task and administered it to approximately one half ($n = 23$) of our English group who had never had instruction in French. The results (see Table 4) do not display the pattern of statistical significance and non-significance found with learner populations: the acceptance rate for speech time answers to discontinuous interrogatives (22.28%) is indistinguishable from the rate of acceptance of speech time answers to continuous interrogatives (30.98%) ($t(22) = 1.59, p = .126$). Past time answers to discontinuous and continuous

Table 4 Native English group ($n = 23$): percentage ‘yes’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation	Past time interpretation
Continuous	30.98 (41.43)	86.95 (20.80)
Discontinuous	22.28 (27.94)	86.95 (28.32)

⁵ A similar response pattern is observable on the basis of ‘no’ responses (see Appendix 4): the French native group ($n = 30$) rejected answers corresponding to the speech time evaluation of discontinuous adjectival restrictions at the rate of 95.00%; whereas they rejected answers corresponding to the speech time evaluation of continuous adjectival restrictions at the rate of 87.08% ($t(29) = 1.71, p = .097$). They rejected answers corresponding to the past time evaluation of discontinuous adjectival restrictions at the rate of (3.75%) and answers corresponding to the past time evaluation of continuous adjectival restrictions at the rate of (10.83%) ($t(29) = 2.04, p = .051$).

interrogatives were accepted at 86.95% ($t(22) = .00, p = 1.000$).⁶ The results of the learner groups could not be achieved simply on the basis of a glossing strategy. Still, the English evidence (for its own sake) suggests a small but nevertheless theoretically significant asymmetry, which confirms the correctness of the claim that the interpretive properties under investigation do not require direct evidence.

As we have seen, the past time interpretation of the adjectival restriction seems preferred so that (native) respondents very systematically find it difficult to consider the less salient interpretation.

The distribution of responses as reflected in the standard deviation measures also seems to be very revealing. In the case of the native French respondents, the SD measures for the discontinuous and continuous constructions differ greatly. For the continuous construction, SD measures show a wide distribution across the two answers: 27.46 and 27.36 for 'yes' and 'no' responses to the speech time answers and 20.06 and 19.89 for 'yes' and 'no' responses to the past time answers. This contrasts with the SD measures for the discontinuous construction, which show a much narrower response distribution: 7.76 and 7.76 for 'yes' and 'no' responses to the speech time answer and 9.93 and 9.93 for 'yes' and 'no' responses to the past time answer. In short, the French natives' response pattern is narrowly distributed with discontinuous interrogatives, while it is more diffuse with continuous interrogatives. This effect can be attributed to ambiguity. Respondents may seize on one reading and reject a possible answer because it does not conform to their expectations. Results are more narrowly distributed with the discontinuous construction, because of the lack of ambiguity. The data from the means support this: the response patterns for answers corresponding to present and past evaluations of adjectival restrictions in continuous (i.e., 12.50% + 88.75% = 101.25) and discontinuous interrogatives (i.e., 5% + 96% = 101) achieve (near) linear dependency. This clearly suggests that, in the cases of the ambiguous discontinuous interrogatives, respondents tend to seize upon one reading and

⁶ A similar response pattern is observable on the basis of 'no' responses (see Appendix 4): the English group ($n = 23$) rejected answers corresponding to the speech time evaluation of discontinuous adjectival restrictions at the rate of 69.02%; whereas they rejected answers corresponding to the speech time evaluation of continuous adjectival restrictions at the rate of 63.04%. This difference of means is not statistically significant ($t(22) = 1.14, p = .268$). There was no statistically significant difference in the rejection of answers corresponding to the past time evaluation of discontinuous adjectival restrictions (4.89%) and their rejection of answers corresponding to the past time evaluation of continuous adjectival restrictions (6.52%) ($t(22) = .83, p = .418$).

provide an answer on the basis of it.

In this respect, the English results stand in sharp contrast. For the native English group, SD measures are similar across the two constructions: For the continuous construction, 41.43 and 42.55 for 'yes' and 'no' responses to speech time answers and 20.80 and 8.31 for 'yes' and 'no' responses to past time answers; and for the discontinuous construction, 27.94 and 35.12 for 'yes' and 'no' responses to speech time answers and 28.32 and 8.20 for 'yes' and 'no' responses to past time answers. In other words, the distinction in the response distribution by construction found in the native French group is certainly not in evidence in the behaviour of the native English group. This is clearly suggestive of a difference between the French and English groups with respect to the licensing of adjectival restrictions. Still, in the response pattern of the English group, we find a slight skewing suggestive of UG principles. The behaviour can be readily understood in terms of the modularity of linguistic knowledge and associated performance systems: the language-dependent parameterized lexicon is independent of the UG principles. Respondents attempt sentence parses constrained by the principles of UG. However, these parses are incompatible with the parameter values of the English functional lexicon. The higher dispersal levels measured by the SD values (which we interpret as uncertainty) come from the fact that these parses are not licensed by (i.e., violate) the parameter values of the English functional lexicon.⁷

Turning to the advanced learners, SD rates are affected by the construction type (this holds for both learner groups). For the continuous construction, SD rates are 37.95 and 40.14 for 'yes' and 'no' responses to the speech time answer and 26.96 and 23.67 for 'yes' and 'no' responses to the past time answer. For the discontinuous construction, SD rates are 19.43 and 21.18 for 'yes' and 'no' responses to the speech time answer and 14.88 and 11.67 for 'yes' and 'no' responses to the past time answer. These SD rates are parallel to those obtained with the French native speakers in being affected by the construction type.

⁷ An anonymous reviewer asks whether results from the French and English groups are comparable, since the task asks what ungrammatical sentences mean for the English group, while the sentences are grammatical for the French group. But here the reviewer is assuming (at some level) what it is that we are trying to demonstrate: that there is a difference in the intuitive interpretability of the experimental sentences in English grammars vs. English–French interlanguage (and French) grammars. However, such a difference has to be empirically demonstrated.

V Mental design vs. judicious strategies

As we have just seen, the results of the learner groups could not be achieved simply on the basis of a glossing strategy.⁸ But what if learners are more judicious in their choices, and use intuitions arising from (10) and (11) (see above) instead? Results given in Table 5 are for a task in all respects similar to the task above, except that it did not involve the partitive particle *of*. It was administered to the remaining half ($n = 24$) of our English group with no history of French instruction. The acceptance rate for speech time answers to discontinuous interrogatives (28.12%) is statistically different from their acceptance of speech time answers to continuous adjectival interrogatives (5.73%) ($t(23) = 2.73, p = .012$). There was no statistically significant difference in the acceptance of past time answers to continuous (89.06%) and discontinuous (93.75%) interrogatives ($t = 1.30, p = .205$).⁹ This is also clearly suggestive that

Table 5 Native English group ($n = 24$): percentage ‘yes’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation	Past time interpretation
Continuous	28.12 (39.39)	89.06 (23.69)
Discontinuous	5.72 (9.73)	93.75 (11.05)

⁸ An anonymous reviewer points out that there are cases of (grammatical) PP modification in English yielding the relevant asymmetry (i)–(ii).

- i) Who in the Conservative Party smoked (cannabis) in the 1970s?
- ii) Who smoked (cannabis) in the Conservative Party in the 1970s?

The reviewer raises the question of whether learners could arrive at the French paradigm by analogical extension to grammatical (i)–(ii), ‘if any substance can be given to that notion’. Extensionally, it is very odd to think that learners (in the usual sense of individuals) would reliably compare with precisely this construction, rather than opt for a ‘glossing’ strategy. From an intensional point of view, the interpretive asymmetry in (i)–(ii) receives an analysis as above: This asymmetry is a reflex of the human language computational system (C_{HL}). Analogical extension here means that the new string is derived by C_{HL} just like (i) and (ii), a point which does not hinge on the particulars of the theory. Furthermore, the evidence gathered in these experiments shows acquisition to be severely constrained. This is precisely what is expected if significant aspects of both L1 and L2 acquisition are required by the structure of mental architecture along the lines investigated by Minimalism. We do not know of any other way in which a mentally feasible theory of analogical extension can be obtained without significant aspects of the model considered here (i.e., a universal set of constraints/principles inherent in mental organization supporting (domain-specific) deduction).

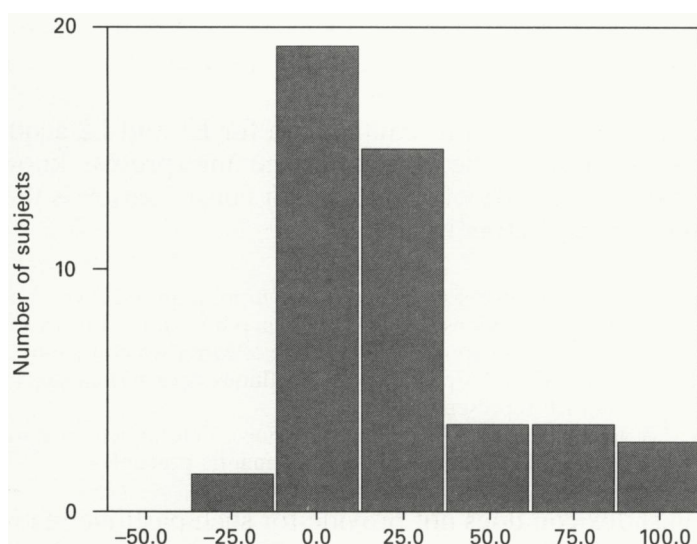
⁹ A similar response pattern is observable on the basis of ‘no’ responses (see Appendix 4): the English group ($n = 24$) rejected answers corresponding to the speech time evaluation of discontinuous adjectival restrictions at the rate of 71.04%, whereas they rejected answers corresponding to speech time evaluation of continuous adjectival restrictions at the rate of 92.18%. This difference achieves significance ($t(23) = 2.46, p = .022$). There was no statistically significant difference in the rejection of answers corresponding to past time evaluation of continuous adjectival restrictions (3.64%) and their rejection of answers corresponding to past time evaluation of continuous adjectival restrictions (10.42%) ($t(23) = 1.92, p = .067$).

aspects of interpretation must clearly pre-date input. However, it still does not directly account for learner behaviour. If these English 'judgements' were the source of the IL intuitions, we would expect higher resolution results than those we actually find in IL.

These empirical results raise the question of whether IL intuitions result from the direct effect of IL hypotheses and principles rooted in mental design or whether respondents simply call on their English intuitions perhaps in a variety of ways. In the latter case, the lower effects would be due to the level of judiciousness of the strategies that different learners employ. But we would naturally expect a high level of idiosyncrasy in how individual learners approach the learning problem: we expect a spectrum of response strategies, with some learners adopting more successful strategies than others.

On the other hand, if the asymmetries that we have observed in IL are reflexes of UG, then a subject distribution chart should show a very strong skewing in just the expected direction. Indeed, the subject distribution chart of the acceptance by our intermediate population of speech time answers to continuous interrogatives vs. discontinuous interrogatives (percentage 'yes' to present interpretation of continuous *de*-restrictions vs. percentage 'yes' to present interpretation of discontinuous *de*-restrictions) shows evidence for two groups of learners (see Figure 5). The horizontal axis charts the difference in the acceptance of continuous vs. discontinuous *de*-restrictions. The vertical axis charts the number of subjects. The subject distribution in Figure 5 shows a clear skewing in the expected direction. This is unexpected under the strategy view, because different learner strategies entail a freedom of behaviour that is clearly absent. Such a distribution with only two outliers with negative values suggests that the hypothesis search space is highly restricted, as predicted if the L2 linguistic data must be treated in domain-specific ways as a function of grammatical organization.

This clearly shows that accounting for IL behaviour in terms of a broad range of strategies is highly suspect. Another issue that must be reckoned with is the fact that there appears to be growth, since advanced learners have a narrower dispersal rate and a lower acceptance rate of speech time answers to discontinuous interrogatives than intermediate learners. In the context of strategies, this growth is curious at best. This would suggest that advanced learners use what English intuitions they have more wisely than intermediate learners. Why they would do precisely that (and not the opposite) is, however, a mystery, since the input is of no direct teaching value in this respect. In the absence of effective



Difference: the acceptance of continuous vs. discontinuous *de*-restrictions

Figure 5 Percentage ‘yes’ to present interpretation of continuous *de*-restrictions vs. percentage ‘yes’ to present interpretation of discontinuous *de*-restrictions
Notes: SD = 32.70; mean = 17.1; *n* = 47.00

input why there is in fact such a skewing of responses and apparent growth is mysterious.

Following the logic of Dekydtspotter *et al.* (1997), there are two ways to construe the hypothesis that knowledge of the interpretive contrast is in some sense ‘simply’ derivative of the English L1 grammar: as an extensional hypothesis and as an intensional hypothesis. On the extensional hypothesis, the performance of the English–French respondents results from English intuitions about English glosses of the French sentences in the task. As we have just seen, the extensional hypothesis fails to provide an account for IL behaviour. The intensional strategy offers grammatical development from the principles of computations at work in the grammar of English. However, recall that on current Minimalist understanding of grammar, there is no principle of computation relevant to the TL, but not to the L1, because the grammar of any natural language is the result of a language-dependent parameterized lexicon, universal principles of grammar allowing just certain representations, and a universal syntax–semantics interface determined (in part) by syntactic and semantic principles. There is no sense in which the grammar of English lacks the universal principles with which the French-like parameterization

interacts. Once the English–French learner has adopted the relevant properties of the French functional lexicon, the interpretive properties follow.

To see precisely what this entails both for L1 and L2 acquisition, we need to consider the source of the interpretive knowledge. Recall that the property of French under consideration is the result of the interacting factors in (12).

- 12) a. A lexical hypothesis on which the *wh*-pronoun *qui* ('who') admits of partitive restrictions and the *wh*-expression *qui* ('who') can equally satisfy the *wh*-feature of C as *qui de célèbre* ('which famous person').
- b. General principles preventing superfluous derivational steps between PF and LF representations.
- c. A universal syntax–semantics interface, determined in part by the principles in (12b) and in part by semantic principles.

The English lexicon does not provide for such partitive restrictions, hence a parse of the input sentence will not be licensed (i.e., they will be violations) under the English-like hypotheses. Nevertheless, aspects of interpretation follow if the parse is constrained by the UG principles, although not licensed by the English-like IL functional lexicon. English–French IL can be (tentatively) characterized in two steps:

- 1) an earlier state with an English-like IL lexicon, yielding at best a marginal asymmetry resulting from mental design, because the relevant parse is not licensed by (i.e., violates) the interlanguage grammar, followed by:
- 2) a later state in which a French-like functional lexicon has been adopted, with the asymmetries surfacing as the result of (more efficient processing accompanying) the development of the licensing *de* morphology in response to exposure to sequences of the type Q (...) *de* AP.

VI Conclusion

This study argues that (aspects of) interpretive knowledge associated with certain sentence types follow in primary and nonprimary language acquisition alike from 'general principles rooted in the specific character of the human mind' (Chomsky, 1972: 102). If this is the case, such aspects should be pervasive. The evidence suggests that they are: knowledge of the relevant constraints on the interpretation of adjectival restrictions of *wh*-quantifiers with respect to tense appears to arise in the absence of any directly relevant input. In particular, we have adduced

evidence that interpretive knowledge in English–French IL is narrowly regimented, much beyond what can be plausibly extracted from very haphazard input and judicious problem-solving strategies (including those that employ English intuitions). The most reasonable interpretation of the results of this study is that knowledge of interpretation in IL is a reflex of the design of the language faculty.

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Appendix 1 Test items

1. *sportif* 'athletic' quadruple

Context for all four items:

Ken and Jeff were inseparable during their school days. Every afternoon during the coffee break they would get together to chat and gossip. In those days, Ken was very athletic and played on the varsity squash team, but Jeff was a gawky teenager who preferred the library to the gym. They have both undergone radical changes as they approach their 40th birthdays. Now, Ken is a beer-swilling couch potato, whereas Jeff is in terrific shape and almost lives at the gym.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
 Mme Goyette: Qui de sportif bavardait—pendant la pause café—à l'école?
 Élève: Ken
- b. past time answer with discontinuous interrogative
 Mme Goyette: Qui bavardait de sportif—pendant la pause café—à l'école?
 Élève: Ken
- c. speech time answer with continuous interrogative
 Mme Goyette: Qui de sportif bavardait—pendant la pause café—à l'école?
 Élève: Jeff
- d. speech time answer with discontinuous interrogative
 Mme Goyette: Qui bavardait de sportif—pendant la pause café—à l'école?
 Élève: Jeff

Translation of interrogatives:

'What athletic person chatted—during the coffee break—at school?'

2. *célèbre* 'famous' quadruple

Context for all four items:

Attitudes toward smoking have changed drastically since the 1960s. In the 1960s many people would go to bars and smoke every night. For example, Herman the Hermit was a famous rock star in those days and was often seen at bars smoking with Linda Tripp, who was then totally unknown. How times have changed! How it is Linda Tripp who is generally famous, and neither of them smokes any more.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
Mme Goyette: Qui de célèbre fumait—au bistro—dans less années 60?
Élève: Herman the Hermit
- b. past time answer with discontinuous interrogative
Mme Goyette: Qui fumait de célèbre—au bistro—dans less années 60?
Élève: Herman the Hermit
- c. speech time answer with continuous interrogative
Mme Goyette: Qui de célèbre fumait—au bistro—dans less années 60?
Élève: Linda Tripp
- d. speech time answer with discontinuous interrogative
Mme Goyette: Qui fumait de célèbre—au bistro—dans less années 60?
Élève: Linda Tripp

Translation of interrogatives:

‘What famous person smoked—in bars—in the ’60s?’

3. *riche* ‘rich’ quadruple

Context for all four items:

Annabelle and Claudette have been very close friends ever since they met in April 1992. Although Annabelle was fabulously wealthy and Claudette was very poor back then, they spent that summer singing together in opera houses all over Europe. In 1997 their individual fortunes changed radically. Annabelle’s company went bankrupt, and now she has no money at all. On the other hand, Claudette discovered oil in her back yard, and now she is a billionaire.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
Mme Goyette: Qui de riche chantait—en Europe—en 1992?
Élève: Annabelle
- b. past time answer with discontinuous interrogative
Mme Goyette: Qui chantait de riche—en Europe—en 1992?
Élève: Annabelle
- c. speech time answer with continuous interrogative
Mme Goyette: Qui de riche chantait—en Europe—en 1992?
Élève: Claudette

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- d. speech time answer with discontinuous interrogative

Mme Goyette: Qui chantait de riche—en Europe—en 1992?

Élève: Claudette

Translation of interrogatives:

‘What rich person sang—in Europe—in 1992?’

4. *blonde* ‘blonde’ quadruple

Context for all four items:

Margaux and Hélène are very fond of changing their hair color. Last week Margaux was a blonde and Hélène’s was a redhead. Also last week, they both had terrible colds and coughed all the time in class. After recovering from their colds on Saturday, they celebrated by dyeing their hair. Now Hélène is blonde and Margaux has red hair.

Interrogative–response pairs:

- a. past time answer with continuous interrogative

Mme Goyette: Qui de blonde toussait—en classe—la semaine dernière?

Élève: Margaux

- b. past time answer with discontinuous interrogative

Mme Goyette: Qui toussait de blonde—en classe—la semaine dernière?

Élève: Margaux

- c. speech time answer with continuous interrogative

Mme Goyette: Qui de blonde toussait—en classe—la semaine dernière?

Élève: Hélène

- d. speech time answer with discontinuous interrogative

Mme Goyette: Qui toussait de blonde—en classe—la semaine dernière?

Élève: Hélène

Translation of interrogatives:

‘What blonde person coughed—in class—last week?’

Note: The interrogatives in this particular quadruple sound a bit odd ‘out of the blue’, because the Adjective *blonde* is unambiguously feminine and *blond* is generally an individual-level predicate. In the context of the narrative above, however, the interrogatives are quite natural.

5. *fatigué* ‘tired’ quadruple

Context for all four items:

In January Jacques and André both lived in Ottawa. Also in January, a mysterious illness struck Canada. It was terrible for Jacques, who was unable to sleep properly, and he was always tired. Luckily for André, he did not get it then. André was always full of energy. Now Jacques and André live in Montréal. Again a mysterious illness is striking Canada. This time, Jacques does not have a problem with being tired, but oddly enough, André is now tired all the time.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
Mme Goyette: Qui de fatigué habitait—à Ottawa—en janvier?
Élève: Jacques
- b. past time answer with discontinuous interrogative
Mme Goyette: Qui habitait de fatigué—à Ottawa—en janvier?
Élève: Jacques
- c. speech time answer with continuous interrogative
Mme Goyette: Qui de fatigué habitait—à Ottawa—en janvier?
Élève: André
- d. speech time answer with discontinuous interrogative
Mme Goyette: Qui habitait de fatigué—à Ottawa—en janvier?
Élève: André

Translation of interrogatives:

‘What tired person lived—in Ottawa—in January?’

6. *triste* ‘sad’ quadruple

Context for all four items:

Last Thursday was a lovely day. That’s why Madeleine and Georgette’s mother allowed both girls to play with a ball in the garden, even though Madeleine was very sad. Georgette felt fine last Thursday, but now she is sad, and Madeleine is happy. Since it has rained every day this week, both girls have had to stay inside.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
Mme Goyette: Qui de triste jouait—avec un ballon—jeudi passé?
Élève: Madeleine

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- b. past time answer with discontinuous interrogative

Mme Goyette: Qui jouait de triste—avec un jardin—jeudi
passé?

Élève: Madeleine

- c. speech time answer with continuous interrogative

Mme Goyette: Qui de triste jouait—avec un jardin—jeudi
passé?

Élève: Georgette

- d. speech time answer with discontinuous interrogative

Mme Goyette: Qui jouait de triste—avec un jardin—jeudi
passé?

Élève: Georgette

Translation of interrogatives:

‘What sad person played—with a ball—last Thursday?’

7. *aimable* ‘likeable’ quadruple

Context for all four items:

Lucrèce was very friendly back in 1994, but she always ate lunch all by herself in the northwest corner of the cafeteria, because she needed time to read during the day. On the other hand, no one liked Monique back in 1994, because she was rude and cruel. Monique was shunned by the others and was forced to eat lunch all by herself in the southeast corner of the cafeteria. In the last five years, Monique has changed her ways, and people have come to genuinely love her. She now has many, many friends, and they join her for lunch. In the meantime, people have discovered how two-faced Lucrèce really is. Everybody hates Lucrèce, who is now openly rude to everyone. Her presence at the lunch table is tolerated, but not cherished.

Interrogative–response pairs:

- a. past time answer with continuous interrogative

Mme Goyette: Qui d’aimable déjeunait—toute seule—en 1994?

Élève: Lucrèce

- b. past time answer with discontinuous interrogative

Mme Goyette: Qui déjeunait d’aimable—toute seule—en 1994?

Élève: Lucrèce

- c. speech time answer with continuous interrogative

Mme Goyette: Qui d’aimable déjeunait—toute seule—en 1994?

Élève: Monique

- d. speech time answer with discontinuous interrogative

Mme Goyette: Qui déjeunait d’aimable—toute seule—en 1994?

Élève: Monique

Translation of interrogatives:

‘What likeable person ate lunch—all alone—in 1994?’

8. *réservé* ‘reserved’ quadruple

Context for all four items:

Twenty years ago, Auguste and Marc-Aurel were students with a problem. Back then, Auguste was very quiet and shy, whereas Marc-Aurel was extremely friendly and out-going. Despite this difference in their personalities, they both were unable to cope with the stress of exams, and they both got completely drunk before every examination. They both came very close to failing out of the university. This afternoon they met at a 20-year college reunion. How times have changed! Now it is Marc-Aurel who is quiet and shy, while Auguste has become completely extroverted, the life of every party. They compared notes and discovered that both of them had sought help for their abusive drinking. They are now both teetotalers.

Interrogative–response pairs:

- a. past time answer with continuous interrogative
Mme Goyette: Qui de réservé s’enivrait—avant chaque examen—il y a 20 ans?
Élève: Auguste
- b. past time answer with discontinuous interrogative
Mme Goyette: Qui s’enivrait de réservé—avant chaque examen—il y a 20 ans?
Élève: Auguste
- c. speech time answer with continuous interrogative
Mme Goyette: Qui de réservé s’enivrait—avant chaque examen—il y a 20 ans?
Élève: Marc-Aurel
- d. speech time answer with discontinuous interrogative
Mme Goyette: Qui s’enivrait de réservé—avant chaque examen—il y a 20 ans?
Élève: Marc-Aurel

Translation of interrogatives:

‘What reserved person got drunk—before each exam—20 years ago?’

Appendix 2 Training instructions

Welcome to the world of Mme Goyette. Mme Goyette teaches English to seventh grade pupils at le Lycée du Cancan in Caen, France. You have been assigned to serve as one of the many judges for the listening comprehension section of the final examination in Mme Goyette's English course. Here we present you with some random examination transcripts. For each item, you will read the English story that the pupil heard. Then follow the actual question (in French) that Mme Goyette asked and the response that the pupil gave. (We have kept the names of the pupils confidential!) For each item you are to indicate whether the pupil has responded with a correct answer on the basis of the English story.

It is important for you to realize that some of Mme Goyette's questions (in the context of the English story) may well have more than one correct answer. The pupil is required to provide just one correct answer, and you should give the pupil credit as long as he or she provides a correct answer. Perhaps some warm-up examples (with questions in English) will be useful to you.

Examples (E1)–(E3)

Pat, Kim, and Lee were sitting in the dormitory lounge stuffing their faces with pizza and discussing their plans for Spring Break. Pat and Kim both said that they were looking forward to their trip to Miami. Pat said, 'I can hardly wait to spend all day lying on the sand and going swimming!' Kim echoed the thought, 'Me, too!' With a shake of the head and a smile, Lee said, 'I'll never understand you guys. I'm going to spend Spring Break right here in Bloomington, enjoying the peace and quiet of the library! You can have Florida!'

Example (E1)

Mme Goyette: Who was looking forward to going to the beach?

The Pupil: Pat

DID THE PUPIL PROVIDE A CORRECT ANSWER? YES NO CANNOT DECIDE

Example (E2)

Mme Goyette: Who was looking forward to going to the beach?

The Pupil: Kim

DID THE PUPIL PROVIDE A CORRECT ANSWER? YES NO CANNOT DECIDE

Example (E3)

Mme Goyette: Who was looking forward to going to the beach?

The Pupil: Lee

DID THE PUPIL PROVIDE A CORRECT ANSWER? YES NO CANNOT DECIDE

You will probably agree that the pupils in examples (E1) and (E2) have provided correct answers. That is, it would be correct to say that Pat was looking forward to going to the beach, and it would be correct to say that Kim was looking forward to going to the beach. Thus, you will no doubt have indicated 'yes' on the answer sheet for examples (E1) and (E2). However, the pupil in example (E3) has clearly given an incorrect answer, since Lee was NOT looking forward to going to the beach. Thus we expect you to have checked 'no' on the answer sheet for example (E3).

Let us turn now to a slightly trickier situation.

Examples (E4)–(E5)

Bill is a fashion photographer, and Harry and Pete are professional models. Yesterday morning Bill and Harry worked on an advertising campaign for a swimwear company. Harry felt cold standing around Bill's air-conditioned studio all morning in nothing but swim trunks. Bill was glad to be wearing a sweater. Yesterday afternoon, Bill and Pete worked on an advertising campaign for a company that makes sunglasses. Since they were working at the beach, Bill wore just his swim trunks, but the art director required Pete to wear a heavy woolen jacket the entire afternoon. Pete was very uncomfortable.

Example (E4)

Mme Goyette: Who did Bill photograph wearing swim trunks yesterday?
The Pupil: Harry

DID THE PUPIL PROVIDE A CORRECT ANSWER? YES NO CANNOT DECIDE

Example (E5)

Mme Goyette: Who did Bill photograph wearing swim trunks yesterday?
The Pupil: Pete

DID THE PUPIL PROVIDE A CORRECT ANSWER? YES NO CANNOT DECIDE

You will no doubt have checked 'yes' for example (E4), since 'Harry' is a correct answer to Mme Goyette's question: Bill photographed Harry wearing swim trunks yesterday. (This can mean that Harry was the one wearing swim trunks.) But 'Pete' in example (E5) is also a correct answer to Mme Goyette's question: Bill photographed Pete wearing swim trunks yesterday. (Notice that this can mean that Bill was the one wearing swim trunks.) Thus, you should have checked 'yes' for example (E5) as well.

Appendix 3

Table 1d Number of responses per continuous interrogative–answer pairing (by group, total population)

	Native speakers				Advanced learners				Intermediate learners			
	yes	no	cd	nr	yes	no	cd	nr	yes	no	cd	nr
CPast 1	27	3	0	0	11	0	0	0	43	3	0	0
CPast 2	27	3	0	0	8	2	0	0	40	6	0	0
CPast 3	27	3	0	0	9	1	1	0	41	5	0	0
CPast 4	28	2	0	0	9	1	1	0	40	6	0	0
CPast 5	29	1	0	0	5	5	1	0	42	4	0	0
CPast 6	23	6	1	0	11	0	0	0	44	2	0	0
CPast 7	26	4	0	0	7	2	2	0	39	7	0	0
CPast 8	26	3	1	0	9	1	1	0	41	5	0	0
CPres 1	7	23	0	0	4	7	0	0	14	31	1	0
CPres 2	6	24	0	0	9	2	0	0	27	16	3	0
CPres 3	2	28	0	0	4	7	0	0	14	32	0	0
CPres 4	2	28	0	0	4	5	2	0	24	21	1	0
CPres 5	3	27	0	0	5	6	0	0	12	33	1	0
CPres 6	4	26	0	0	5	5	1	0	21	25	0	0
CPres 7	5	25	0	0	5	5	1	0	18	28	0	0
CPres 8	3	26	1	0	5	5	1	0	20	25	1	0

Note: cd = cannot decide; nr = no response; CPast = Continuous/past answer; CPres = Continuous/present (speech time) answer

Table 2d Number of responses per discontinuous interrogative–answer pairing (by group, total population)

	Native speakers				Advanced learners				Intermediate learners			
	yes	no	cd	nr	yes	no	cd	nr	yes	no	cd	nr
DPast 1	28	2	0	0	11	0	0	0	40	5	1	0
DPast 2	28	2	0	0	9	2	0	0	23	2	1	0
DPast 3	28	2	0	0	11	0	0	0	44	1	1	0
DPast 4	29	1	0	0	9	1	1	0	43	2	0	1
DPast 5	28	2	0	0	10	0	1	0	42	3	1	0
DPast 6	30	0	0	0	10	1	0	0	41	5	0	0
DPast 7	30	0	0	0	10	1	0	0	41	4	1	0
DPast 8	27	3	0	0	10	0	1	0	40	6	0	0
DPres 1	3	27	0	0	2	7	2	0	7	37	2	0
DPres 2	0	30	0	0	6	13	1	0	16	28	2	0
DPres 3	1	29	0	0	2	8	1	0	7	39	0	0
DPres 4	0	30	0	0	0	11	0	0	10	36	0	0
DPres 5	1	29	0	0	0	10	1	0	6	40	0	0
DPres 6	3	27	0	0	1	10	0	0	7	38	1	0
DPres 7	2	28	0	0	2	9	0	0	15	30	1	0
DPres 8	2	28	0	0	4	6	1	0	18	34	0	0

Note: cd = cannot decide; nr = no response; DPast = Discontinuous/past answer; DPres = Discontinuous/present (speech time) answer

Appendix 4

Table 1a English–French Interlanguage (intermediate learners) ($n = 47$): percentage ‘no’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation*	Past time interpretation
Continuous	56.76 (34.39)	9.57 (21.04)
Discontinuous	73.40 (23.69)	7.98 (15.06)

Note: *This distinction is significant ($t(46) = 3.51$, $p = .001$).

Table 2a English–French Interlanguage (advanced learners) ($n = 11$): percentage ‘no’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions (SD in brackets)

	Speech time interpretation	Past time interpretation
Continuous	48.45 (40.14)	13.63 (23.67)
Discontinuous	80.68 (21.18)	5.68 (11.67)

Note: *This distinction is significant ($t(10) = 2.44$, $p = .035$).

Table 3a Native French group ($n = 30$): percentage ‘no’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions

	Speech time interpretation	Past time interpretation
Continuous	87.08 (27.36)	10.83 (19.89)
Discontinuous	95.00 (7.76)	3.75 (9.93)

Table 4a Native English group ($n = 23$): percentage ‘no’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions

	Speech time interpretation*	Past time interpretation
Continuous	63.04 (42.55)	6.52 (8.31)
Discontinuous	69.02 (35.12)	4.89 (8.20)

Table 5a Native English group ($n = 24$): percentage ‘no’ on speech time and past time interpretations of discontinuous vs. continuous adjectival restrictions

	Speech time interpretation*	Past time interpretation
Continuous	71.04 (39.27)	10.42 (23.51)
Discontinuous	92.19 (11.55)	3.64 (9.38)

Note: *This distinction is significant ($t(23) = 2.46$, $p = .022$).